

Analog-to-Digital and Digital-to-Analog Converters and Circuits

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Lecture References:

- ▶ Book:

- ▶ *Microprocessor Architecture, Programming and Applications with 8085 (Chapter-13)*, **Author:** Ramesh Gaonkor

- ▶ Lecture Materials:

- ▶ *Digital-to-Analog Analog-to-Digital, Interface Part IV, Microprocessor, Georgia State University, Department of Physics and Astronomy.*
 - ▶ *Programming the 8051 Microcontroller*, Dr. Konstantinos Tatas , Embedded Real-Time Processor Systems - Frederick University.
 - ▶ *Analog to Digital Converters*, Byron Johns, Danny Carpenter, Stephanie Pohl, Harry “Bo” Marr, October 4, 2005.

What is Analog Signals?

Analog signals – directly measurable **quantities** in terms of some other **quantity**.

Examples:

- ▶ **Thermometer** – *mercury* height rises as *temperature* rises.
- ▶ **Car Speedometer** – *Needle* moves farther right as you *accelerate*.
- ▶ **Stereo** – *Volume* increases/decreases as you turn the *knob*.

What is Digital Signals?

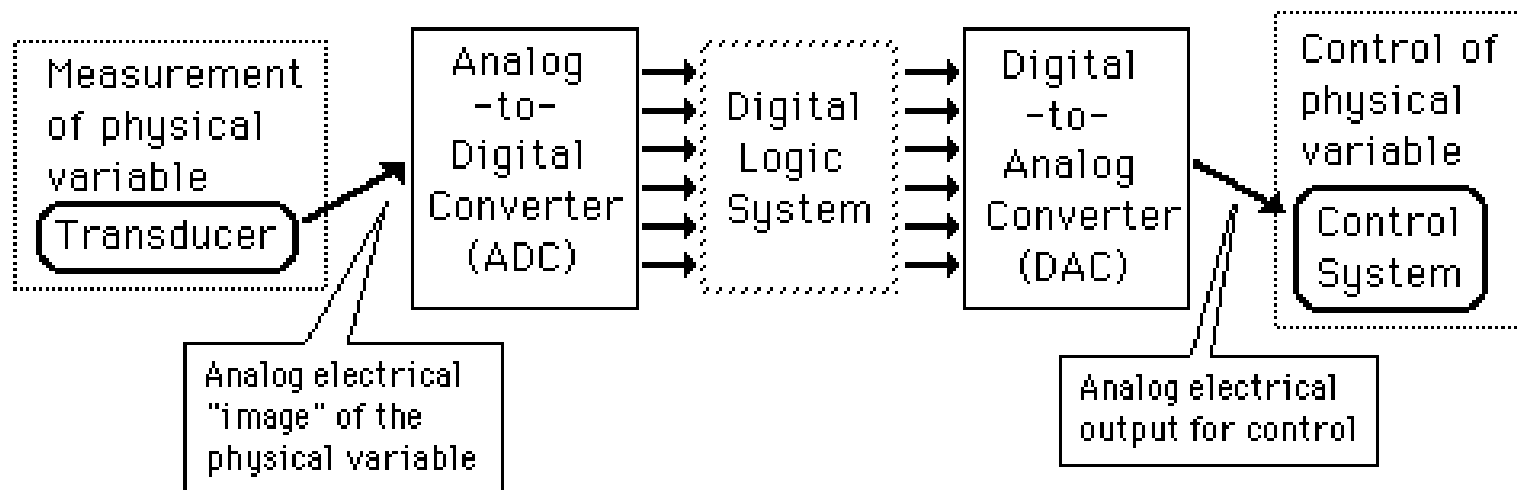
Digital Signals – have only **two** states. For digital computers, we refer to binary states, 0 and 1. “1” can be on, “0” can be off.

Examples:

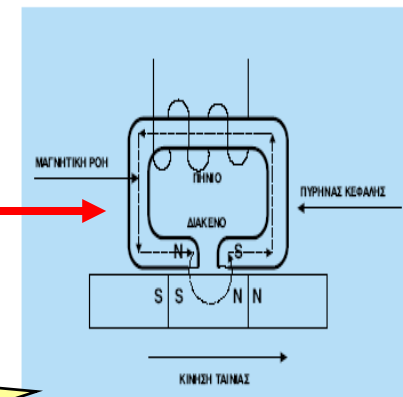
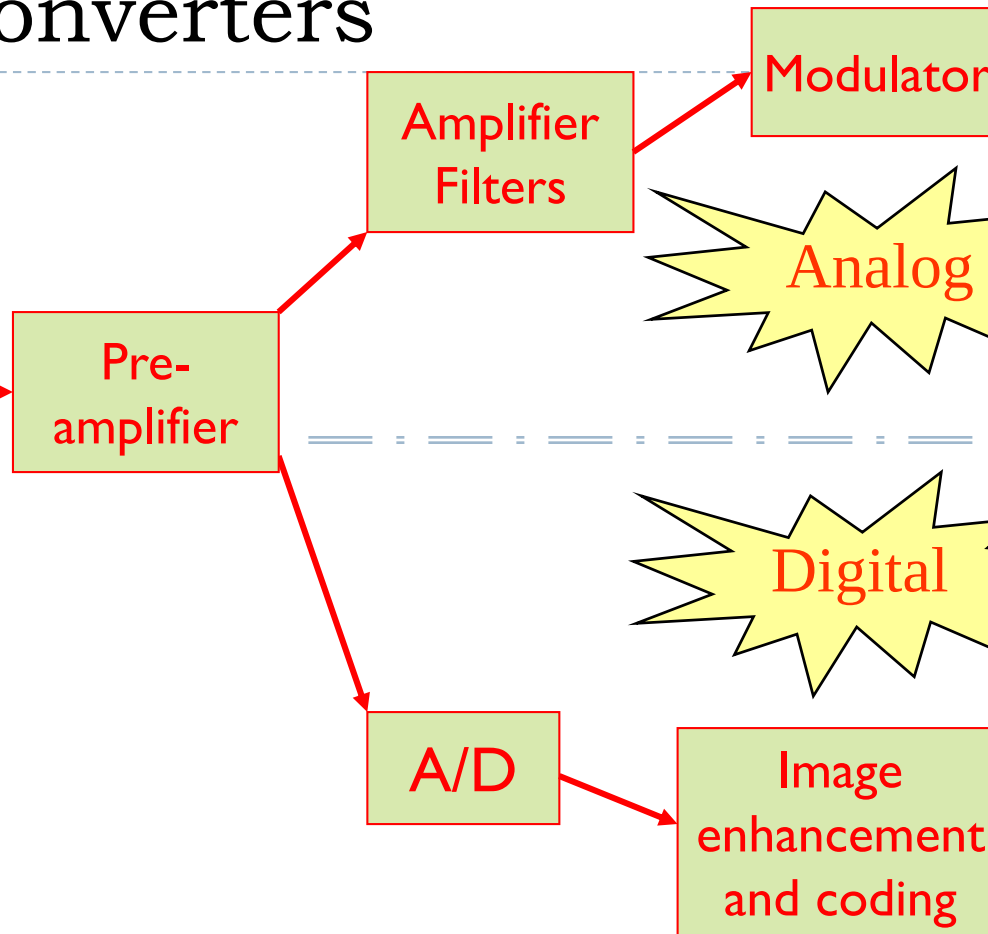
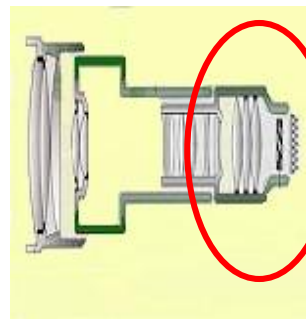
- ▶ Light switch can be either on or off
- ▶ Door to a room is either open or closed

Data Handling System

- ▶ Both data about the physical world and control signals sent to interact with the physical world are typically "analog" or continuously varying quantities.
- ▶ In order to use the *power of digital systems*, one must convert from **analog to digital** form on the experimental measurement end and convert from **digital to analog** form on the control or output end of a laboratory system.



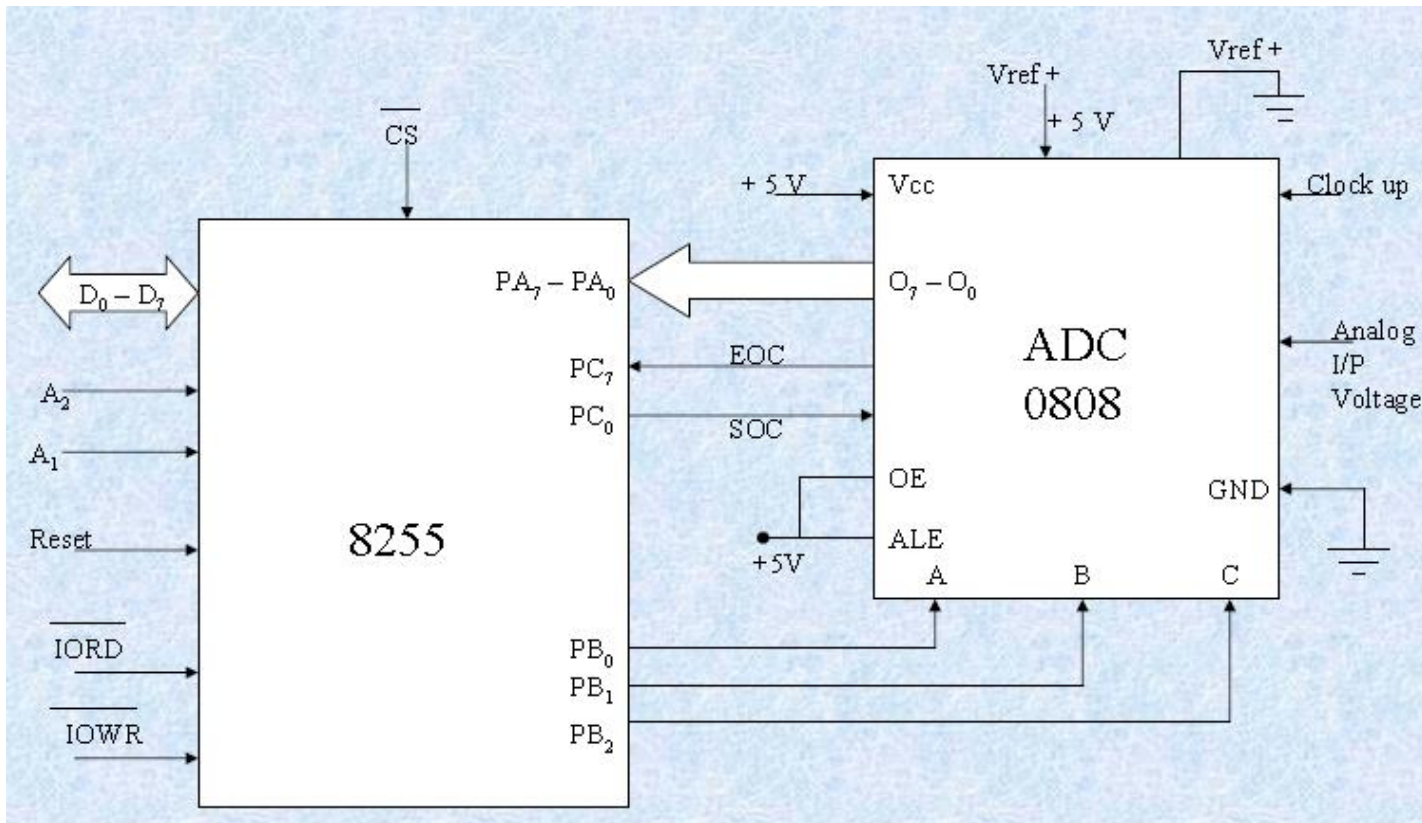
Data Converters



- ▶ **Analog to Digital Converters (ADC):** Convert an analog quantity (voltage, current) into a digital code.
- ▶ **Digital to Analog Converters (DAC):** Convert a digital code into an analog quantity (voltage, current).

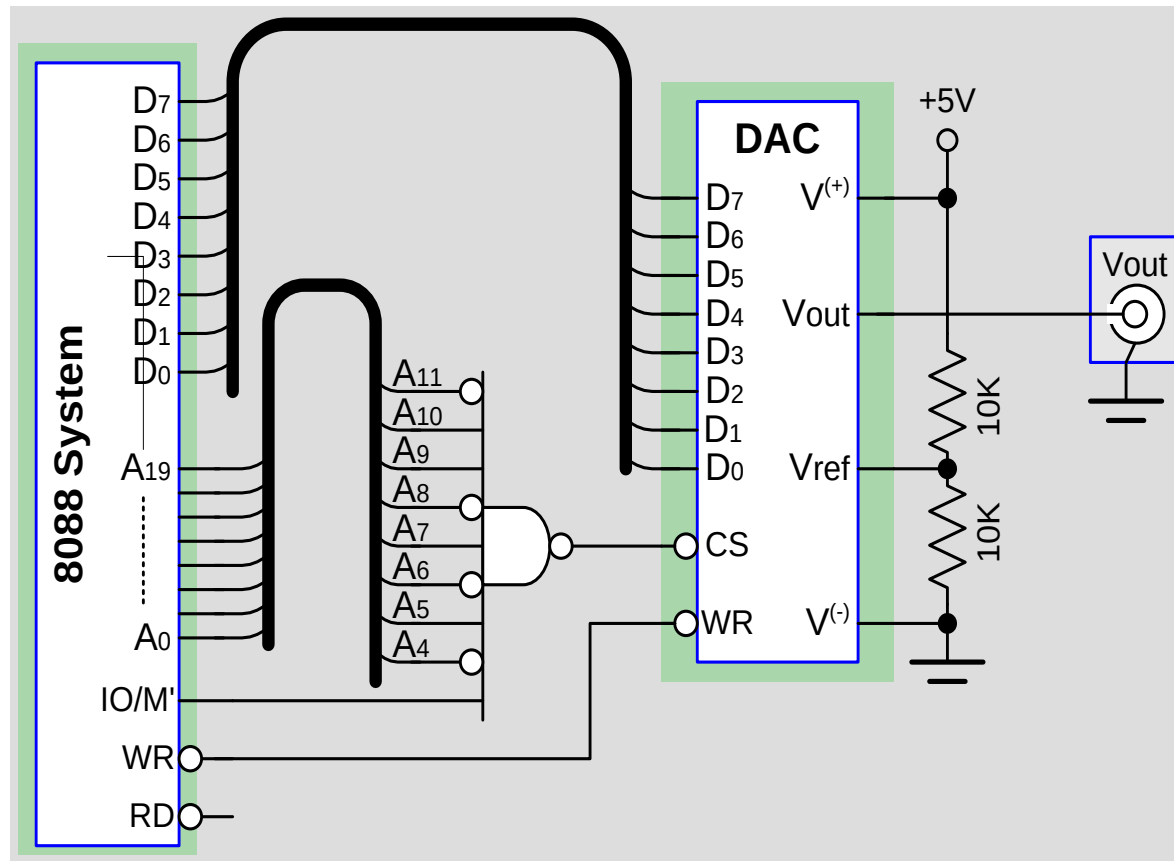
Interfacing with Data Converters

- ▶ Microprocessor compatible data converters are attached with the microprocessor through I/O interfaces.

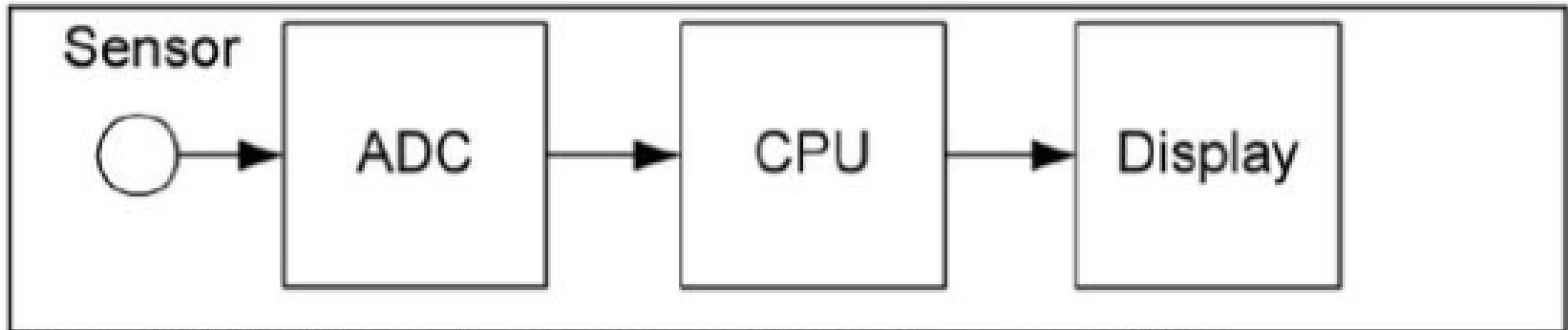


Interfacing with Data Converters

- ▶ Microprocessor compatible data converters are attached on the microprocessor's bus as standard I/O devices.

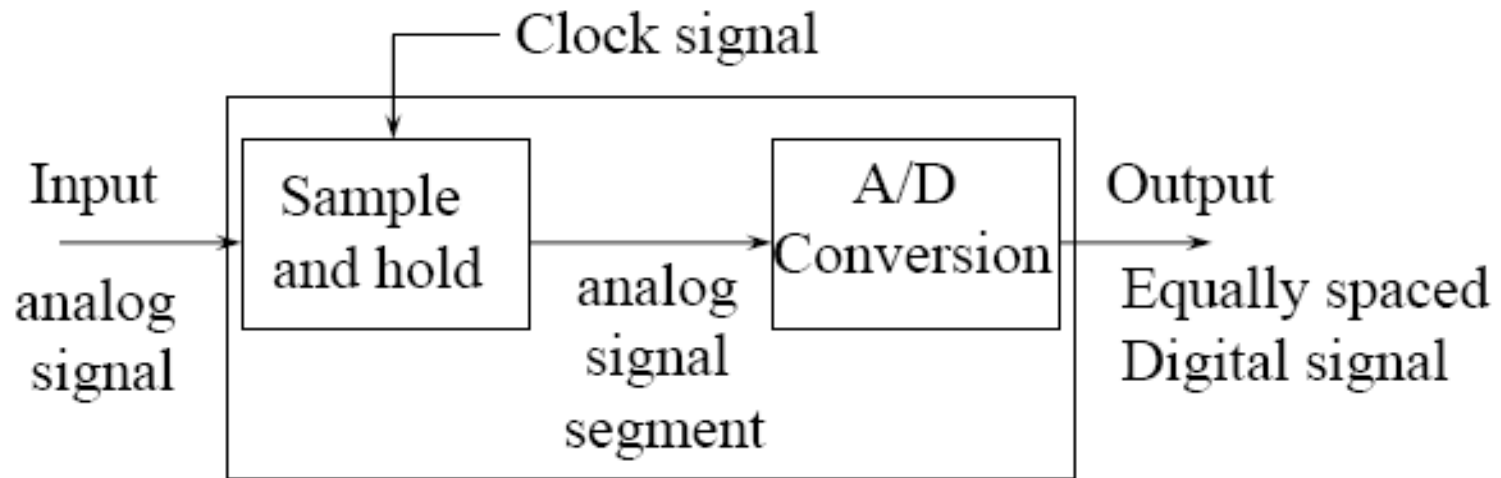


Interfacing with Data Converters and Sensor



What does an A/D converter DO?

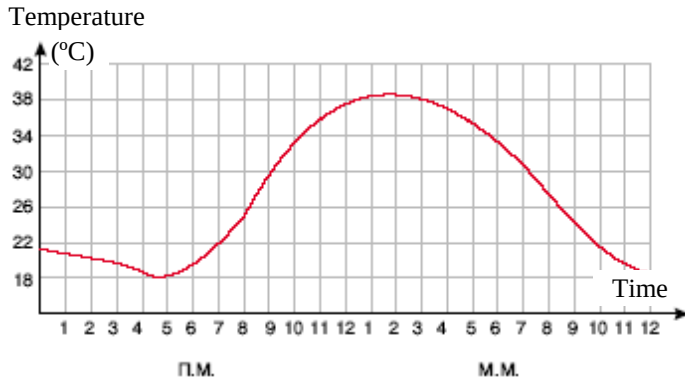
- Converts analog signals into binary words



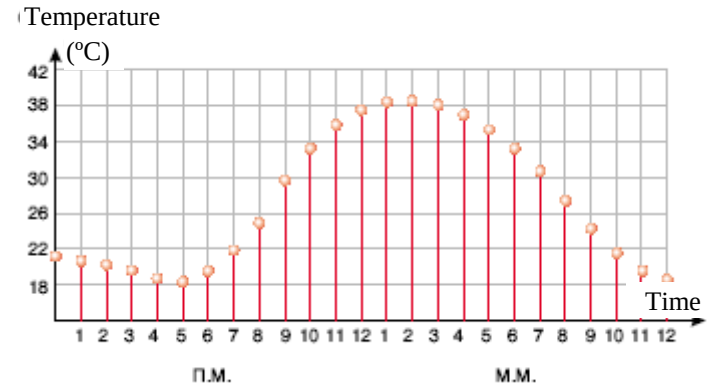
Analog-to-Digital (A/D) Conversion

- ▶ 2-step process required:
 - ▶ **Quantizing** - breaking down analog value is a set of finite states.
 - ▶ **Encoding** - assigning a digital word or number to each state and matching it to the input signal.

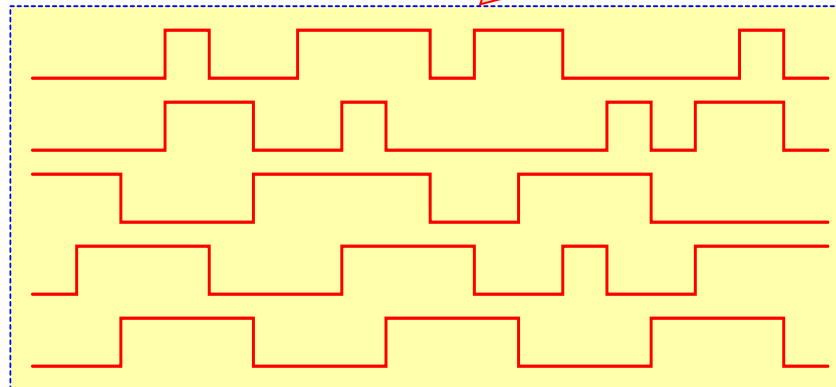
Analog-to-Digital (A/D) Conversion



Sampling & quantization



Coding



Step-1: Quantizing

► Example:

- You have 0-10V signals. Separate them into a set of discrete states with 1.25V increments. (**How did we get 1.25V?**)

Output States	Discrete Voltage Ranges (V)
0	0.00-1.25
1	1.25-2.50
2	2.50-3.75
3	3.75-5.00
4	5.00-6.25
5	6.25-7.50
6	7.50-8.75
7	8.75-10.0

Step-1: Quantizing

- ▶ The number of possible states that the converter can output is:

$$N=2^n$$

where, ***n*** is the number of bits in the A/D converter

- ▶ **Example:** For a 3 bit A/D converter, $N=2^3=8$.
- ▶ **Analog quantization size:**

$$Q=(V_{\max}-V_{\min})/N = (10V - 0V)/8 = 1.25V$$

Step-2: Encoding

- ▶ Assign the digital value (binary number) to each state for the computer to read.

Output States	Output Binary Equivalent
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

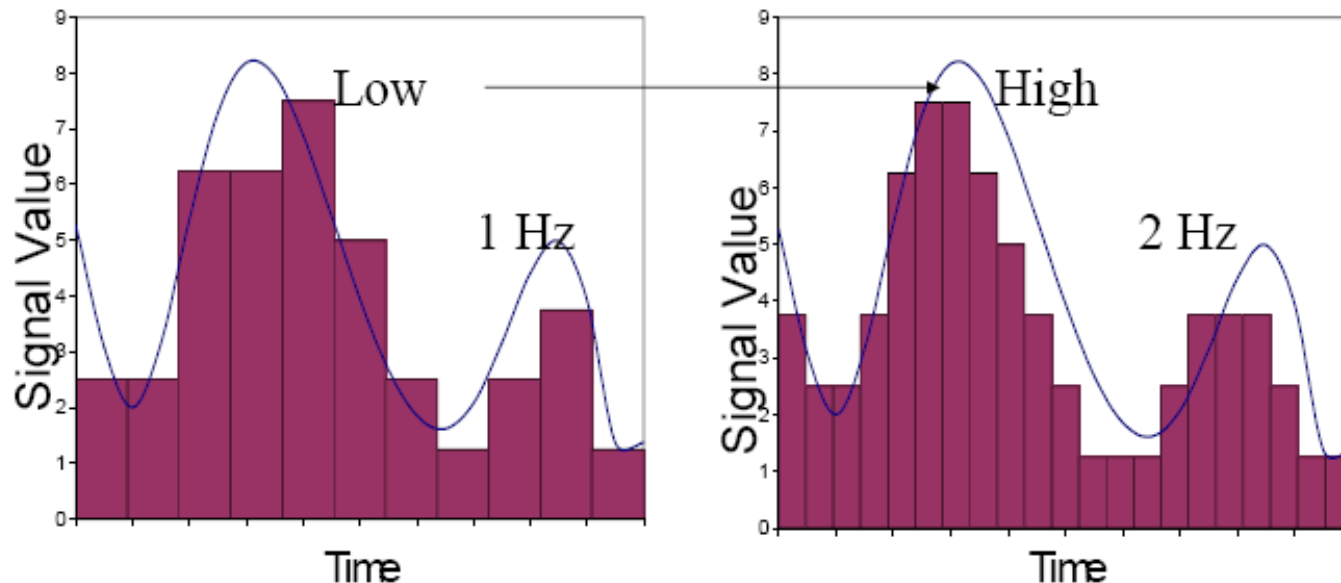
Accuracy of A/D Conversion

- ▶ There are two ways to best improve accuracy of A/D conversion:
 - I. Increasing the **resolution** which improves the accuracy in measuring the amplitude of the analog signal.
 - ▶ **Resolution** (number of discrete values the converter can produce) = Analog Quantization size (Q)
 $(Q) = V_{\text{range}} / 2^n$, where V_{range} is the range of analog voltages which can be represented
 - ▶ limited by **signal-to-noise ratio** (should be around 6dB)
 - ▶ In the previous example: $Q = 1.25\text{V}$, this is a high resolution. A lower resolution would be if we used a 2-bit converter, then the resolution would be $10/2^2 = 2.50\text{V}$.

Accuracy of A/D Conversion

2. Increasing the **sampling rate** which increases the maximum frequency that can be measured.

- ▶ Frequency at which ADC evaluates analog signal. As we see in the second picture, evaluating the signal more often more accurately depicts the ADC signal.

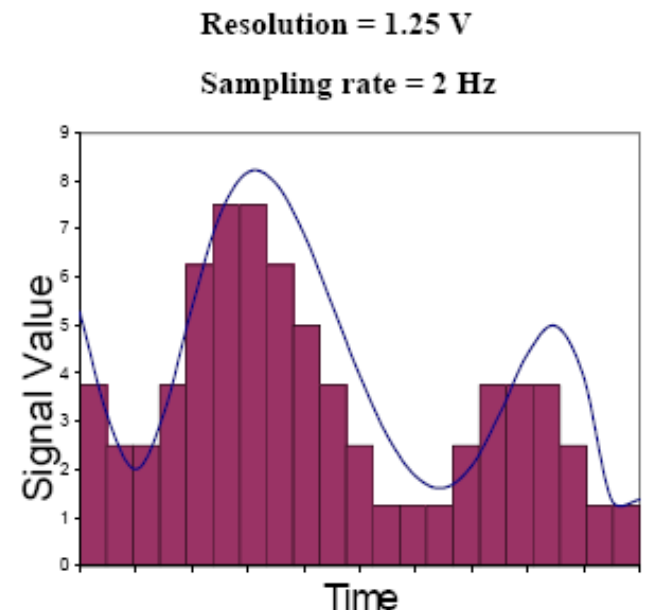
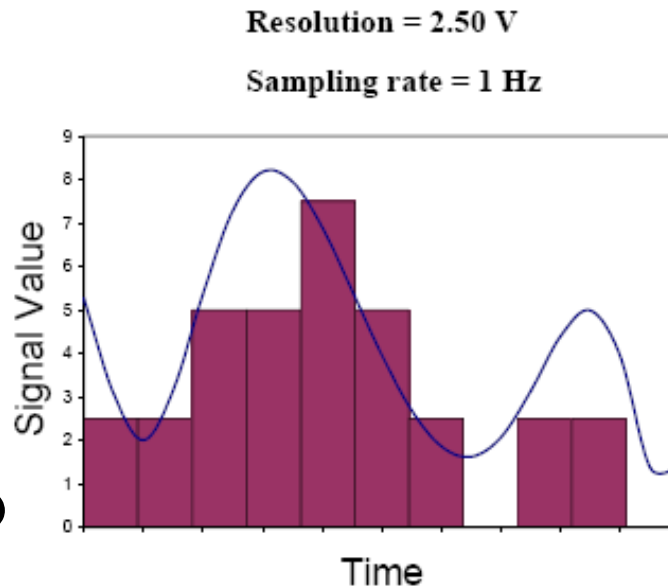


Aliasing Problem

- ▶ Aliasing occurs when the input signal is changing much faster than the sample rate.
- ▶ For example, a 2 kHz sine wave being sampled at 1.5 kHz would be reconstructed as a 500 Hz (the aliased signal) sine wave.
- ▶ **Nyquist Rule:** Use a sampling frequency at least twice as high as the maximum frequency in the signal to avoid aliasing.

Better Accuracy:

Therefore, increasing both the **sampling rate** and the **resolution** you can obtain better accuracy in your AD signals.

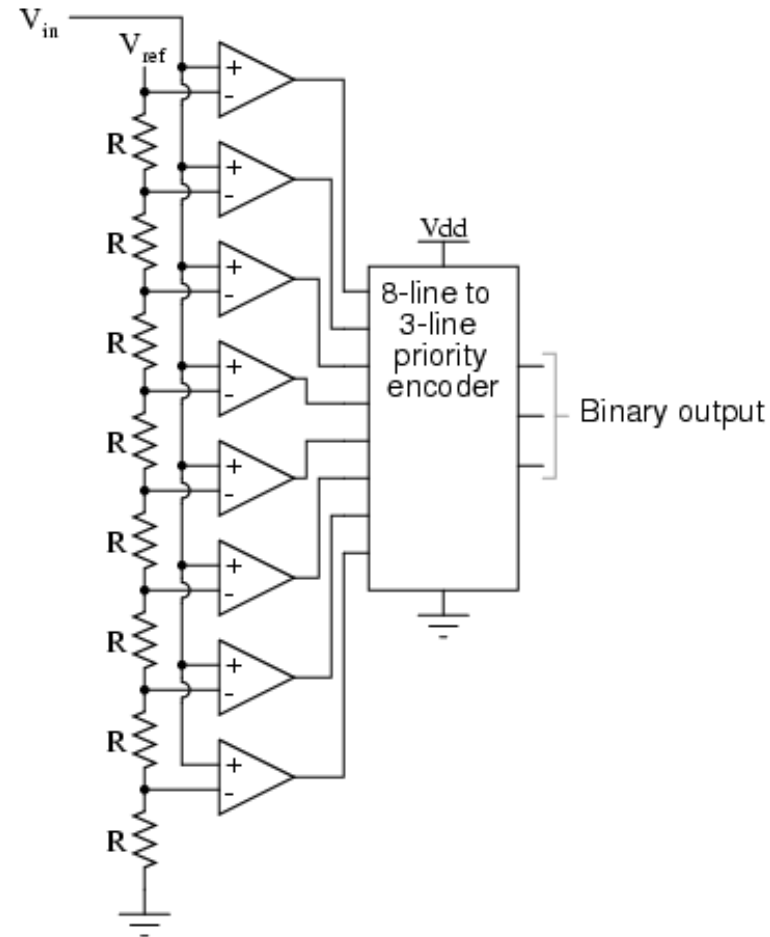


Analog-to-Digital (A/D) Converter Types

- ▶ A/D Converters
 - ▶ Flash ADC
 - ▶ Delta-Sigma ADC
 - ▶ Successive Approximation ADC

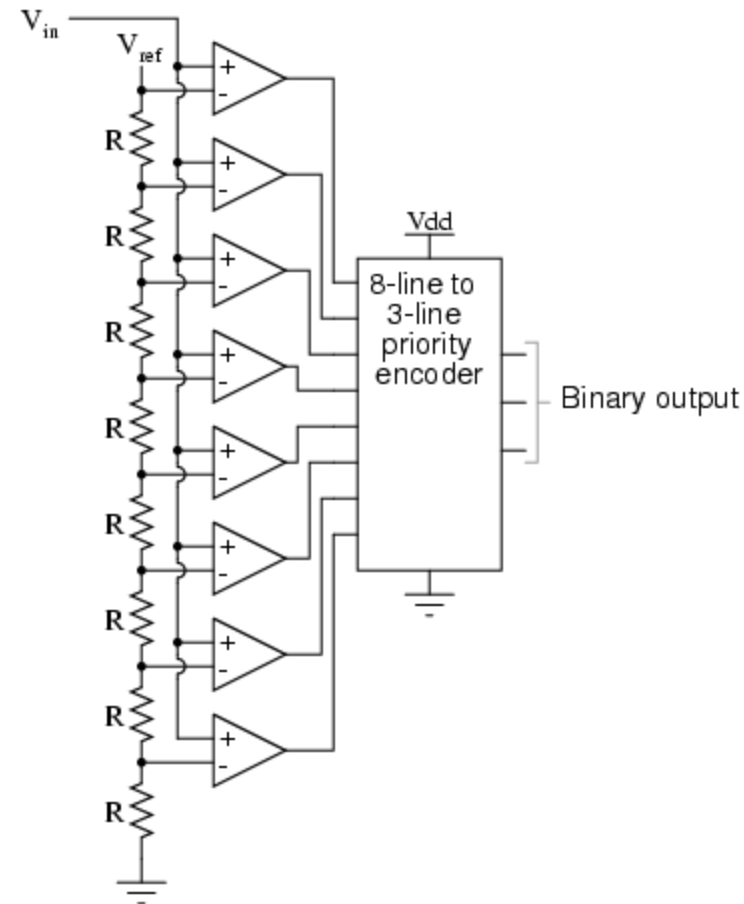
A/D Converters: **Flash ADC**

- ▶ Consists of a series of **comparators**, each one comparing the input signal to a **unique reference voltage**.
- ▶ The comparator outputs connect to the inputs of a **priority encoder** circuit, which produces a binary output
- ▶ As the **analog input voltage exceeds** the reference voltage at each comparator, the comparator outputs will sequentially saturate to a **high** state.
- ▶ The priority encoder generates a binary number based on the highest-order active input, ignoring all other active inputs.



A/D Converters: **Flash ADC**

- ▶ It is the fastest type of ADC available, but requires a comparator for each value of output. (8 for 3-bit, 64 for 6-bit, 256 for 8-bit, etc.)
- ▶ Such ADCs are available in IC format up to 8-bit and 10-bit flash ADCs (1024 comparators) are planned.
- ▶ The encoder logic executes a truth table to convert the ladder of inputs to the binary number output.



A/D Converters: **Flash ADC**

▶ **Advantages**

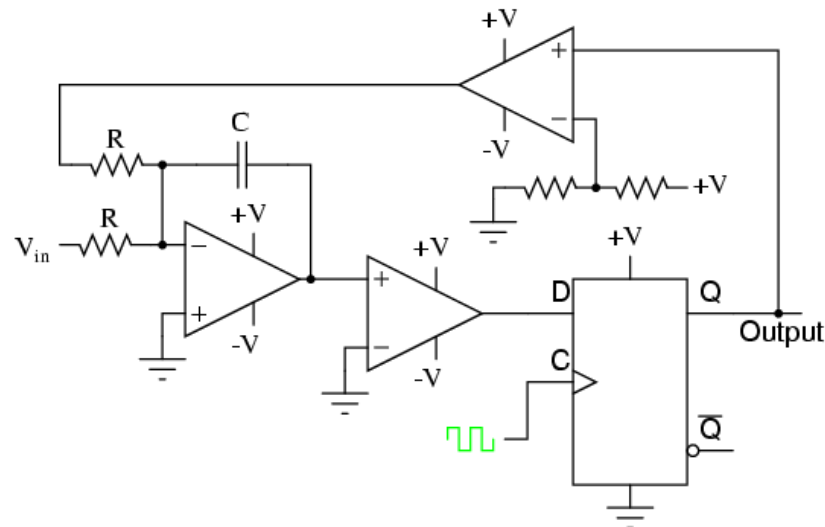
- ▶ Simplest in terms of operational theory
- ▶ Most efficient in terms of speed, very fast

▶ **Disadvantages**

- ▶ Lower resolution
- ▶ Expensive
- ▶ Limited only in terms of comparator and gate propagation delays
- ▶ For each additional output bit, the number of comparators is doubled
 - i.e. for 8 bits, 256 comparators needed

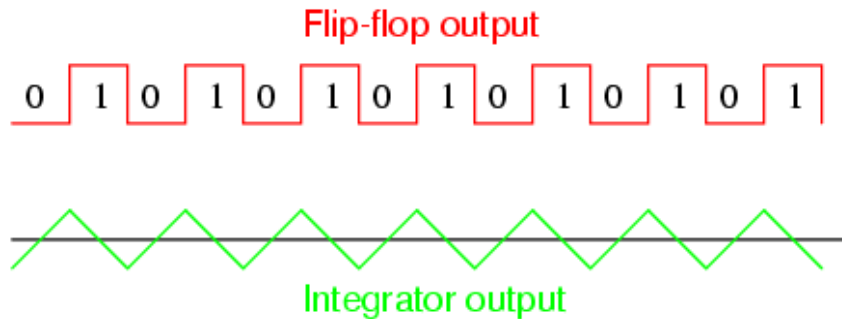
A/D Converters: **Delta-Sigma ADC**

- ▶ It involves changing an analog signal into time or frequency and comparing these parameters with the known values.
- ▶ Over sampled input signal goes to the integrator
- ▶ Output of integration is compared to GND
- ▶ Iterates to produce a serial bit stream
- ▶ Output is serial bit stream with # of 1's proportional to V_{in}

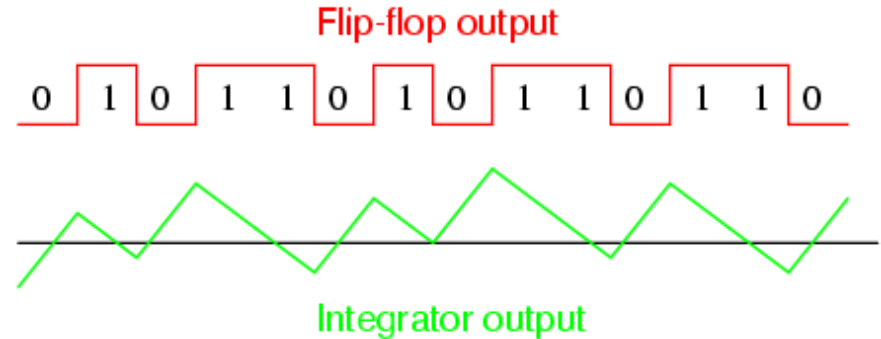


A/D Converters: **Delta-Sigma ADC**

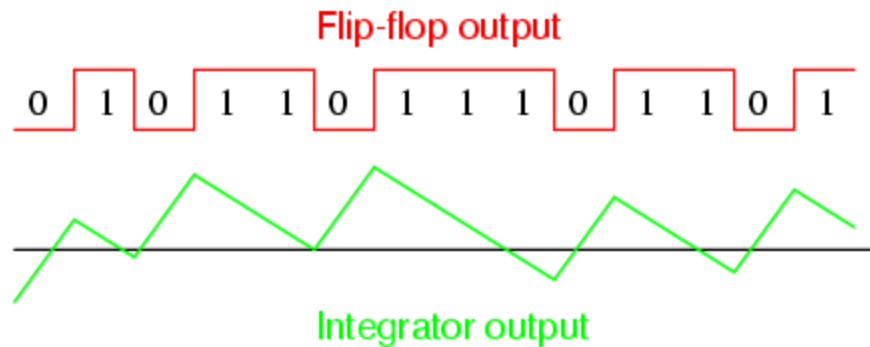
*$\Delta\Sigma$ converter operation with
0 volt analog input*



*$\Delta\Sigma$ converter operation with
medium negative analog input*



*$\Delta\Sigma$ converter operation with
large negative analog input*



A/D Converters: **Delta-Sigma ADC**

▶ **Advantages**

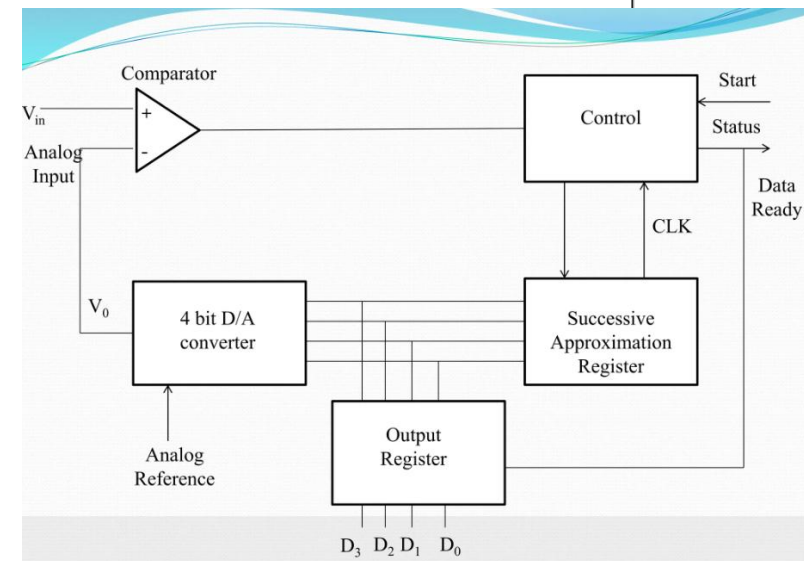
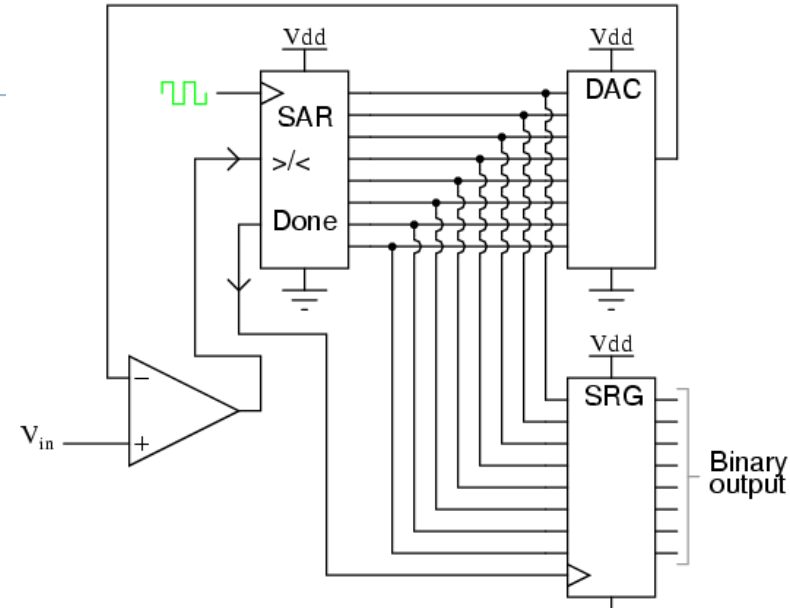
- ▶ High resolution
- ▶ No precision external components needed
- ▶ Accuracy is high

▶ **Disadvantages**

- ▶ Slow due to oversampling

A/D Converters: **Successive Approximation ADC**

- ▶ A Successive Approximation Register (SAR) is added to the circuit.
- ▶ Instead of counting up in binary sequence, this register counts by trying all values of bits starting with the MSB and finishing at the LSB.
- ▶ The register monitors the comparators output to see if the binary count is greater or less than the analog signal input and adjusts the bits accordingly.
- ▶ **Applications:** Data Loggers and Instrumentation.



A/D Converters: **Successive Approximation ADC**

▶ Example:

- ▶ 10 bit resolution or $0.0009765625V$ of V_{ref}
- ▶ $V_{in} = 0.6$ volts
- ▶ $V_{ref} = 1$ volts
- ▶ Find the digital value for V_{in}

▶ Step 1: MSB (Bit 9)

- ▶ Divide V_{ref} by 2 to get V
- ▶ Compare $V = V_{ref}/2$ with V_{in}
- ▶ If V_{in} is greater than V , turn MSB on (1)
- ▶ If V_{in} is less than V , turn MSB off (0)
- ▶ $V_{in} = 0.6V$ and $V = 1/2 = 0.5$
- ▶ Since $V_{in} > V$, MSB = 1 (on) ➡

Bit	Voltage
9	.5
8	.25
7	.125
6	.0625
5	.03125
4	.015625
3	.0078125
2	.00390625
1	.001952125
0	.0009765625

1									
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A/D Converters: **Successive Approximation ADC**

▶ **Step 2:** MSB-1 (Bit 8)

▶ Compare $V_{in}=0.6\text{ V}$ to $V=V_{ref}/2 + V_{ref}/4= 0.5+0.25 =0.75\text{V}$

▶ Since $0.6<0.75$, MSB-1 is turned off (0).

1	0								
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▶ **Step 3:** MSB-2 (Bit 7)

▶ Go back to the last voltage that caused it to be turned on (MSB, Bit 9) and add it to $V_{ref}/8$, and compare with V_{in}

▶ Compare V_{in} with $(0.5+V_{ref}/8)=0.625$

▶ Since $0.6<0.625$, MSB-2 is turned off (0).

1	0	0							
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▶ **Step 4:** MSB (Bit 6)

▶ Go to the last bit that caused it to be turned on (In this case MSB, Bit 9) and add it to $V_{ref}/16$, and compare it to V_{in}

▶ Compare V_{in} to $V= 0.5 + V_{ref}/16= 0.5625$

▶ Since $0.6>0.5625$, MSB-3 turned on (1)

MSB	MSB-1	MSB-2	MSB-3	...					
1	0	0	1						

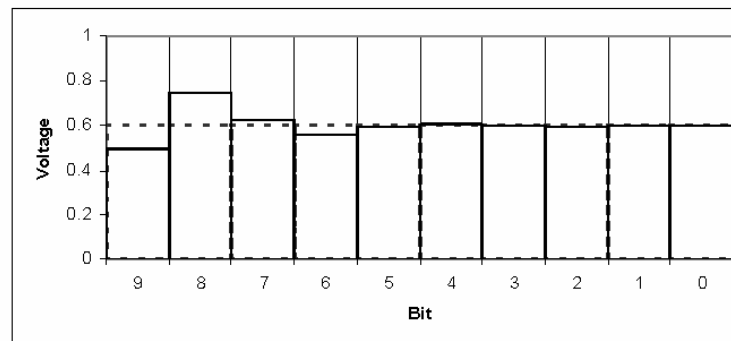
A/D Converters: **Successive Approximation ADC**

- ▶ This process continues for all the remaining bits.

•Digital Results:

MSB	MSB-1	MSB-2	MSB-3	...					LSB
1	0	0	1	1	0	0	1	1	0

•Results: $\frac{1}{2} + \frac{1}{16} + \frac{1}{32} + \frac{1}{256} + \frac{1}{512} = .599609375 \text{ V}$



A/D Converters: **Successive Approximation ADC**

▶ **Advantages**

- ▶ Capable of high speed and reliable.
- ▶ Medium accuracy compared to other ADC types
- ▶ Good tradeoff between speed and cost
- ▶ Capable of outputting the binary number in serial (one bit at a time) format.

▶ **Disadvantages**

- ▶ Higher resolution successive approximation ADC's will be slower.
- ▶ Less accurate than the other converters.

A/D Converters Types Comparision

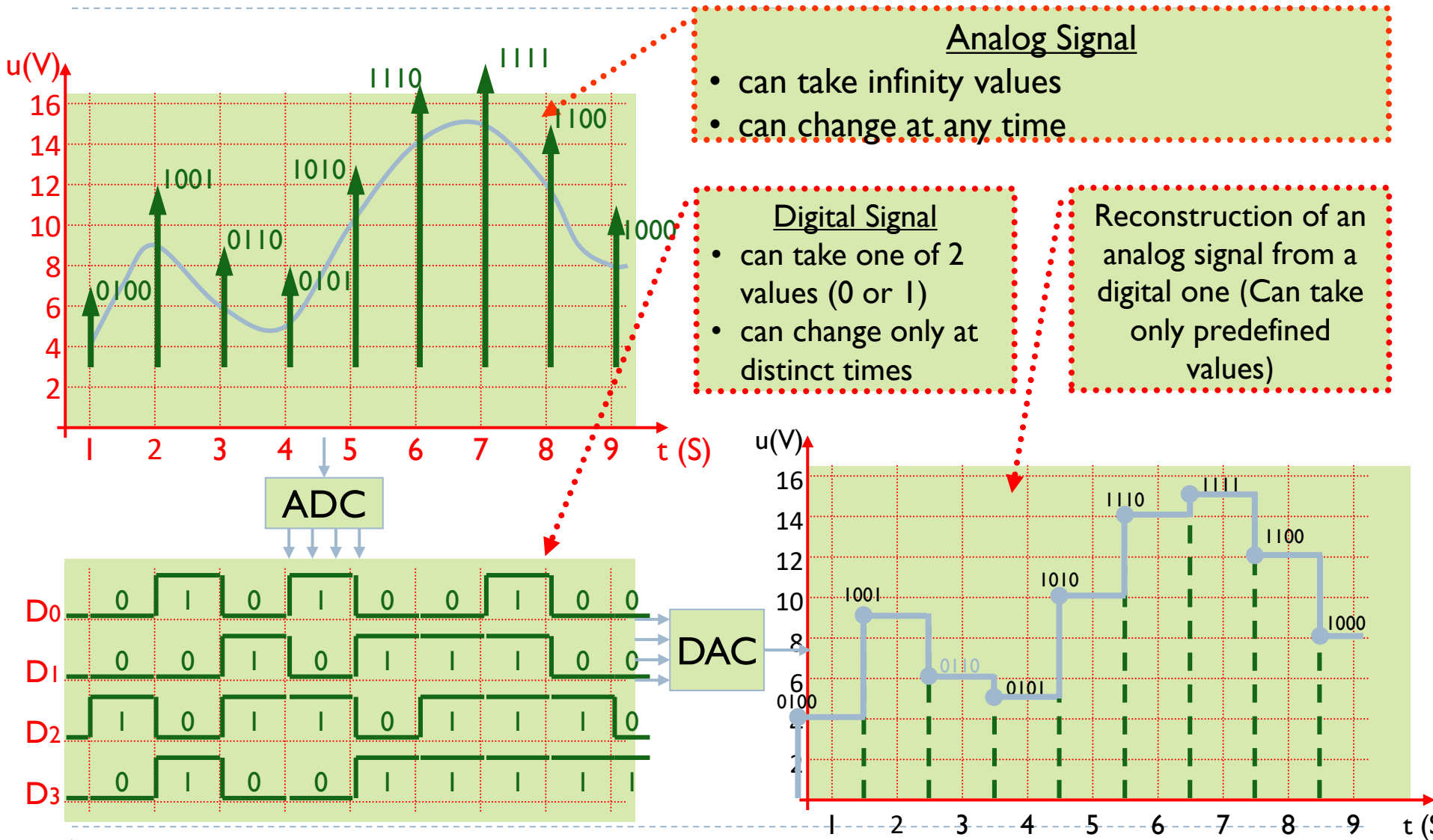
Type	Speed (relative)	Cost (relative)
Flash	Very Fast	High
Delta-Sigma	Slow	Low
Successive Approximation	Medium – Fast	Low

What does a D/A converter DO?

- ▶ The analog signal at the output of a D/A converter is linearly proportional to the binary code at the input of the converter.
 - ▶ If the binary code at the input is **0001** and the output voltage is **5mV**.
 - ▶ If the binary code at the input becomes **1001**, the output voltage will become **45mV**.
- ▶ If a D/A converter has 4 digital inputs then the analog signal at the output can have one out of 16 values.
- ▶ If a D/A converter has N digital inputs then the analog signal at the output can have one out of 2^N values.

D3	D2	D1	D0	Vout (mV)
0	0	0	0	0
0	0	0	1	5
0	0	1	0	10
0	0	1	1	15
0	1	0	0	20
0	1	0	1	25
0	1	1	0	30
0	1	1	1	35
1	0	0	0	40
1	0	0	1	45
1	0	1	0	50
1	0	1	1	55
1	1	0	0	60
1	1	0	1	65
1	1	1	0	70
1	1	1	1	75

What does a D/A converter DO?

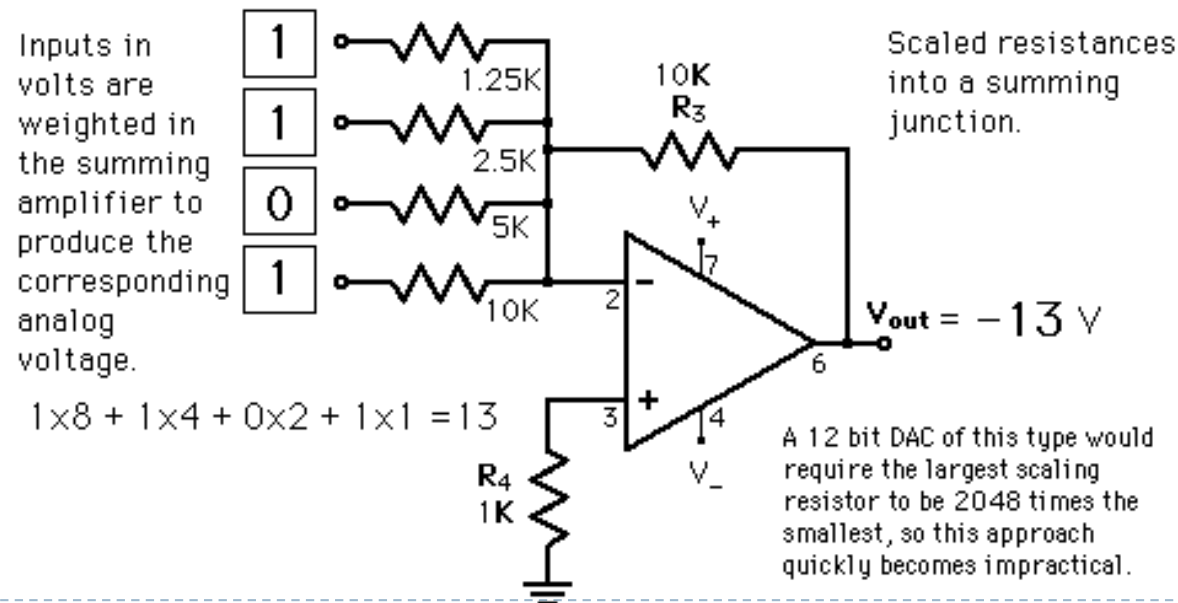


Digital-to-Analog (D/A) Converter Types

- ▶ D/A Converters
 - ▶ Weighted Summing Amplifier
 - ▶ R-2R Network Approach

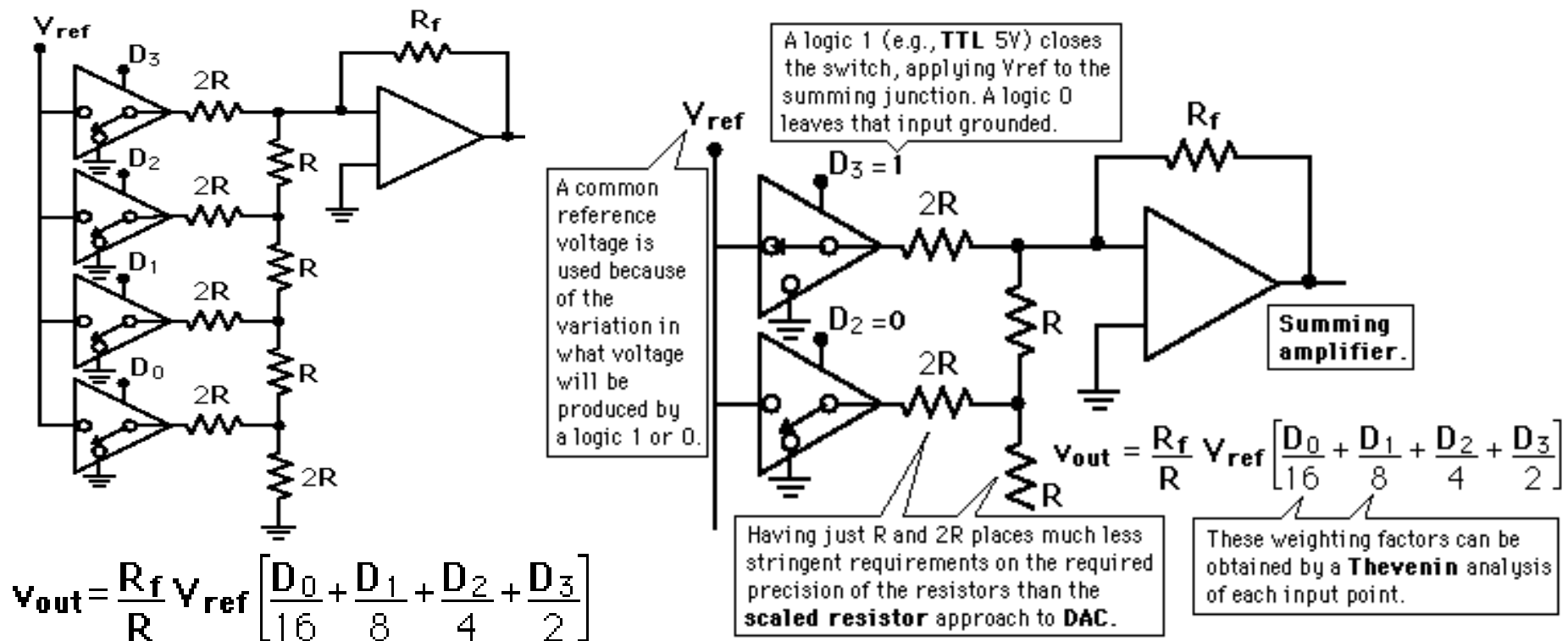
D/A Converters: **Weighted Sum DAC**

- ▶ One way to achieve D/A conversion is to use a **summing amplifier**.
- ▶ This approach is not satisfactory for a large number of bits because it requires too much precision in the summing resistors.
- ▶ This problem can be overcome in the R-2R network DAC.



D/A Converters: **R-2R Ladder DAC**

- ▶ The summing amplifier with the R-2R ladder of resistances produces the output where the D's take the value 0 or 1.
- ▶ This is illustrated for 4 bits, but can be extended to any number with just the resistance values R and 2R.



Thank You !!

